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# Observable versus Latent Factors

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## ABSTRACT

We test for temporal stability in local linear projection coefficients of observable factors on latent ones embedded in a high-dimensional vector obeying a linear factor model. The proposed test explores the fact that, under the null hypothesis, residuals from global linear projections of observable factors on latent ones, computed over a fixed time interval via Principal Component Analysis (PCA), should also be locally uncorrelated with the PCA factors. The test is fully nonparametric. Its asymptotic behavior is derived under a joint infill and large cross-section asymptotic setup. The test can be used to decide whether and over what horizon observable factors can be linearly spanned by latent ones. In an empirical application, we show that a linear relation between the market volatility factor and the latent systematic risk factors embedded in the cross-section of stock returns exists only over short periods of length of one trading day. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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High-frequency data; Latent factor model; Nonparametric test; PCA

## 1. Introduction

Linear factor models are widely used in many fields and in particular in finance for modeling asset returns. One can treat the factors as latent and use conventional Principal Component Analysis (PCA) to extract them (up to a rotation). However, the extracted factors are typically hard to interpret. An alternative approach is to take a stand on the factors and relate them to observable variables. For example, in the case of asset returns, the observable factors will typically be traded portfolios of assets.

In general, there is a gap between the latent factors and the observable ones, as noted by Connor and Korajczyk (1991), Fan, Ke, and Liao (2021), Giglio and Xiu (2021), and Patton and Simsek (2024) in the context of asset returns. This can be due to measurement error or more generally because the observable factors might contain risks that do not generate systematic movements in asset prices. The goal of the current paper is to study the relationship between observable and latent factors. In most applications to date, this relation is assumed to be linear and time-invariant, but this does not need to hold in general. We design a nonparametric test to decide if this is the case.

Specifically, suppose that a high-dimensional vector obeys an approximate linear factor model, with a latent factor being a continuous martingale that we denote with  $f_t$ . Consider a one-dimensional observable factor  $g_t$ . One can decompose the continuous martingale part of  $g_t$  in the following way:

$$\int_0^t \gamma_s^\top df_s + \epsilon_t^g, \quad (1.1)$$

for some process  $\gamma_t$  and a continuous martingale  $\epsilon_t^g$  that is orthogonal in martingale sense to  $f_t$ . The coefficient process

$\gamma_t$  is a local linear projection coefficient at time  $t$  and is time-varying in general. The objective of this article is to test whether  $\gamma_t$  remains constant on a fixed interval  $t \in [0, T]$ . This null hypothesis is satisfied in linear factor models commonly used in applied work.

Our test uses high-frequency observations of the observable factors and the high-dimensional vector, that obeys a linear factor model, over a time interval of fixed length. We estimate the latent factors via PCA, and run a linear regression of the high-frequency increments of the observable factors on the latent ones to collect the residuals from this regression. Under the null hypothesis, these residuals should have zero quadratic covariation with the PCA factors over the entire estimation window. In technical terms, the residuals from the global linear projection should form a martingale that is orthogonal in a martingale sense to the latent factors. Our test statistic is a weighted sum of the product of the estimated residuals and the latent factors. By varying the weights over a class of weight functions, we can determine whether the orthogonality of the residuals and latent factors holds locally at any point in time. Our test, therefore, is based on the supremum of the test statistic over a suitable class of weight functions, which is powerful against the alternatives of interest.

We derive the limiting distribution of the test statistic considered as a function of the weights used in its construction. The limiting distribution of the statistic is nonstandard. To compute the critical values of the test, we develop a wild bootstrap approach based on multiplying each of the residuals from the linear regression of observed factors on latent ones by iid standard normal random variables and recomputing the test statistic using the bootstrapped data. We show that, with this way

of computing the critical values, the test achieves asymptotically the desired size and has an asymptotic power of one.

In the low-frequency setting, a typical approach to testing time-varying coefficients is based on “nonparametric kernel testing”, which consists of consistently estimating  $\gamma_t$  under the alternative and comparing it with a time-invariant estimator when the null hypothesis is imposed, see for example, Härdle and Mammen (1993), Dette (1999), Gao and Gijbels (2008), and Chen and Huang (2018). If the null hypothesis fails to hold, the two estimators should have a significant discrepancy, yielding its rejection. This approach requires modeling the time-variation of  $\gamma_t$  so that a consistent estimator can be derived under the alternative. Another complication, specific to our case of latent factor models and highly relevant from an empirical point of view, is that some latent factors might be dormant at certain points in time, which yields  $\gamma_t$  not identifiable at these time points. Our formulation of the null hypothesis in terms of properties of the residuals from a global linear projection of observable on latent factors allows us to avoid these complications and derive a test with minimal assumptions about the time-series behavior of  $\gamma_t$  and the latent factors.

The current article relates to several strands of work. First, it is connected to earlier work that considers estimation of assets’ exposures toward observable factors from high-frequency data. Contributions in this area include Barndorff-Nielsen and Shephard (2004), Andersen et al. (2006), Mykland and Zhang (2006), Mykland and Zhang (2009), Todorov and Bollerslev (2010), Mancini and Gobbi (2012), Li, Todorov, and Tauchen (2017), Aït-Sahalia, Kalnina, and Xiu (2020) and Chen, Mykland, and Zhang (2024), among many others. Second, our article is also related to earlier work on estimation via PCA analysis of latent factor models from high-frequency data, see for example, Aït-Sahalia and Xiu (2017, 2019), Fan and Kim (2018), Pelger (2019, 2020), Chen, Mykland, and Zhang (2020) and Chen (2024), of conditional latent factor models from low-frequency data, see for example, Zaffaroni (2019), and of models with observed and latent factors together, see for example, Chen, Lu, and Xie (2025). Third, changes over time in the span of latent factor loadings have been studied by Kong (2017), Kong et al. (2023) and Liao and Todorov (2024). Fourth, the relation between observable and latent risk factors in low-frequency setting is studied by Fan, Ke, and Liao (2021), Andreou et al. (2019, 2025) and Fortin, Gagliardini, and Scaillet (2023). Unlike these studies, in our work, we allow for a gap between the latent and the observable factors which is important in applications as noted recently by Giglio and Xiu (2021). Finally, the article is also related to the statistical literature on varying-coefficient models, where the coefficient is either modeled as a smooth function of time or a function of additionally observed variables. Contributions in this literature include Gallant and Fuller (1973), Hastie and Tibshirani (1993), Cai, Fan, and Li (2000), Lin and Ying (2001), Tian, Zucker, and Wei (2005), Ahmad, Leelahanon, and Li (2005), and Wang, Nan, and Kalbfleisch (2024), among others.

The article is organized as follows. Section 2 presents the setup and formulates the testing problem. We develop our test and characterize its asymptotic properties in Section 3. We evaluate its finite-sample behavior in Section 4. Our empirical application is in Section 5. Section 6 concludes. The Appendix in the main text shows that models with time-varying factor

loadings are covered by our setting and provides a motivation for the test for applications in finance. The Supplementary Appendix contains the technical assumptions and the proofs.

## 2. Setup and Formulation of the Testing Problem

### 2.1. The Model

On a standard probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  equipped with the filtration  $(\mathcal{F}_t)_{t \geq 0}$ , we have the following continuous-time linear factor model for the  $p \times 1$  vector  $Y_t$  over a fixed time interval  $[0, T]$ :

$$dY_t = \alpha_t dt + \beta df_t + dJ_t + d\epsilon_t, \quad (2.1)$$

where  $\alpha_t$  is the drift term,  $\beta$  is a matrix of factor loadings,  $f_t$  is a vector of continuous (latent) factors,  $J_t$  is the jump part of  $Y_t$  and  $\epsilon_t$  captures the noise term (with its elements having zero quadratic covariation with each other, apart from a finite number of elements, and with  $f_t$ ).  $\alpha_t$ ,  $J_t$  and  $\epsilon_t$  are  $p \times 1$  vectors,  $f_t$  is  $K \times 1$  vector and  $\beta$  is  $p \times K$  matrix, for some integer  $K$ . Even though the factor loadings matrix is time-invariant, it is easy to show that the above model nests a dynamic model with a finite number of discrete changes in the factor loadings by suitably expanding the factor space. This is shown formally in the Appendix and an example is provided in the Monte Carlo section.

Our interest in the article is one-dimensional observable factor  $g_t$  and its relation with the latent factors  $f_t$ . Towards this end, we can decompose the continuous part of  $g_t$ , denoted by  $g_t^c$ , as

$$g_t^c = \gamma_{t,0} + \int_0^t \gamma_s^\top df_s + \epsilon_t^g, \quad (2.2)$$

for some continuous finite variation process  $\gamma_{t,0}$ , some process  $\gamma_t$ , and a continuous martingale  $\epsilon_t^g$  with  $\langle f, \epsilon^g \rangle_t = 0$ , for  $t \in [0, T]$ , where we denote the quadratic covariation between two arbitrary stochastic processes  $x$  and  $y$  as

$$\langle x, y \rangle_t = \text{plim}_{\Delta \downarrow 0} \frac{1}{\Delta} \sum_{i=1}^{\lfloor t/\Delta \rfloor} (x_{i\Delta} - x_{(i-1)\Delta})(y_{i\Delta} - y_{(i-1)\Delta}). \quad (2.3)$$

The finite variation process  $\gamma_{t,0}$  is the continuous-time analogue of the conditional mean of  $g_t$  in discrete settings. Our interest in this article is in the martingale component of  $g_t$ .

We can view  $\int_0^t \gamma_s^\top df_s$  in the above decomposition as the local linear projection of the martingale component of  $g_t$  on  $f_t$ , see Mykland and Zhang (2006). We can give an explicit expression for  $\gamma_t$ . For this, we denote

$$c_t^{ff} = \frac{d\langle f, f \rangle_t}{dt} \quad \text{and} \quad c_t^{fg} = \frac{d\langle f, g \rangle_t}{dt}, \quad (2.4)$$

which are the spot variance of  $f$  and the spot covariance between  $f$  and  $g$ . With this notation,  $\gamma_t$  is given by

$$\gamma_t = (c_t^{ff})^{-1} c_t^{fg}, \quad (2.5)$$

where in the above, the inverse should be interpreted as a generalized inverse. This can cover the situation in which at certain points in time some of the factors in  $f_t$  are dormant.

## 2.2. Formulation of the Testing Problem

Our goal is to test whether  $\gamma_t$  remains constant on the time interval  $[0, T]$ , which is the case for linear factor models. There are many other interesting questions that can be studied in the above framework. For example, in the context of applications in finance if  $g_t$  and  $Y_t$  are prices of financial assets, then the decomposition in (2.2) separates the risk in  $g_t$  into two parts: one that can be replicated by creating dynamic portfolios of the assets in the vector  $Y_t$  and one that cannot. These objects are of independent interest. We leave studies of estimators for them for future work.

A natural approach to developing a test for the temporal stability of  $\gamma_t$  is to compare local estimates of  $\gamma_t$ , which do not assume that it is constant on  $[0, T]$ , with a global one that assumes that this is the case. Such a test, however, is not easy to develop because some of the factors might not be active, that is, have nonzero variance, on the entire time interval. For this reason, we propose an alternative formulation of the null hypothesis, which provides an easy way of constructing a test for it. Consider the global linear projection of  $g_t$  on  $f_t$  over the interval  $[0, T]$ :

$$g_t^c = \gamma_{t,0} + \gamma^\top f_t + u_t, \quad (2.6)$$

for the same  $\gamma_{t,0}$  as above, some random variable  $\gamma$ , and a continuous martingale process  $u_t$  satisfying  $\langle f, u \rangle_T = 0$ . The random variable  $\gamma$  (adapted to  $\mathcal{F}_T$ ) is the counterpart of the (global) linear projection coefficient in classical regression settings and is given by

$$\gamma = \langle f, f \rangle_T^{-1} \langle f, g \rangle_T \equiv \left( \int_0^T c_t^{ff} dt \right)^{-1} \int_0^T c_t^{fg} dt. \quad (2.7)$$

Using the expression for the local projection coefficients, we can express equivalently  $\gamma$  as a weighted average of these local coefficients:

$$\gamma = \int_0^T \omega_t \gamma_t dt, \text{ where } \omega_t = \left( \int_0^T c_t^{ff} dt \right)^{-1} c_t^{ff}.$$

In general, we do not need to have  $\langle f, u \rangle_t = 0$  for  $t < T$ . On one hand, the part of  $f_t$  of interest is typically  $\int_0^t \gamma_s^\top df_s$  and not  $\gamma^\top f_t$  as it captures all covariation between  $g_t$  and  $f_t$ . This is illustrated in the context of a finance application in the Appendix. On the other hand,  $\gamma$  is much easier to measure from the data than  $\gamma_t$  as the former can use the data from the whole interval  $[0, T]$  while the latter can use only a small fraction of it, mainly the observations in a local neighborhood of  $t$ . To see this, note that an estimator of  $\gamma$  based on time series data in  $[0, T]$  has an asymptotic variance that is proportional to  $T$  in a baseline homoscedastic setting.

Given the above analysis, our null hypothesis is that  $u_t$  is a martingale that is locally uncorrelated with the latent factors  $f_t$  over the whole time interval  $[0, T]$ . That is,

$$\mathbb{H}_0 : \langle f, u \rangle_t = 0, \text{ for } \forall t \in [0, T], \quad (2.8)$$

while our alternative is

$$\mathbb{H}_1 : \langle f, u \rangle_t = c, \text{ for some } t \in [0, T] \text{ and } c \neq 0. \quad (2.9)$$

We will use these formulations of the hypotheses to derive our test in the next section. To see the connection with the time variation in  $\gamma_t$ , note that we can write

$$\langle f, u \rangle_t = \int_0^t (\gamma_s - \gamma)^\top d\langle f, f \rangle_s.$$

Therefore, if  $\gamma_t$  varies over some part of  $[0, T]$ , then the process  $\langle f, u \rangle_t$  will deviate from zero at some point in  $[0, T]$ .

We finish this section with a comment. In this article we focus on the case where  $g_t$  is univariate. However, our framework can be readily extended to the multivariate observable factor case to study questions about the relation between the span of observable factors, once measurement error in them is accounted for, and that of the latent ones. We leave such an extension of the analysis for future work.

## 3. The Test

We now develop the test for the null hypothesis and analyze its asymptotic properties. We start with introducing the observation scheme and constructing the test statistic.

### 3.1. Construction of the Test Statistic

The vector  $Y_t$  and the factor  $g_t$  are observed on the equidistant grid  $0, \frac{T}{n}, \dots, T$ , for  $T$  fixed and  $n \rightarrow \infty$ . We use  $\Delta_n = \frac{T}{n}$  for the mesh of the observation grid. The increments of a generic process  $G$  are denoted with

$$\Delta_i^n G = G_{i\Delta_n} - G_{(i-1)\Delta_n}.$$

To remove the jumps from  $Y_t$ , we will do trimming of the increments following Mancini (2001). More specifically, denote  $h(x, a) = (x \vee -a) \wedge a$ , for some  $x \in \mathbb{R}$  and  $a \in \mathbb{R}_+$ . Then, the scaled and trimmed increments of  $Y_t$  are denoted with

$$\bar{Y}_{t,j} = \frac{h(\Delta_t^n Y, \gamma_j \Delta_n^\varpi)}{\sqrt{\Delta_n}}, \quad \bar{Y} = (\bar{Y}_{t,j} : t = 1, \dots, n, j = 1, \dots, p), \quad (3.1)$$

for some  $\gamma_j > 0$  that is uniformly bounded in  $j$  and a constant  $\varpi \in (0, 1/2)$ .

We introduce similar notation for the scaled increments of the factors and of the noise term:

$$\bar{f}_t = \frac{\Delta_t^n f}{\sqrt{\Delta_n}}, \quad \bar{g}_t = \frac{\Delta_t^n g}{\sqrt{\Delta_n}}, \quad \bar{\epsilon}_t = \frac{\Delta_t^n \epsilon}{\sqrt{\Delta_n}}, \quad t = 1, \dots, n, \quad (3.2)$$

and

$$\bar{F} = (\bar{f}_t : t = 1, \dots, n), \quad \bar{G} = (\bar{g}_t : t = 1, \dots, n), \\ \bar{\mathcal{E}} = (\bar{\epsilon}_{t,j} : t = 1, \dots, n, j = 1, \dots, p). \quad (3.3)$$

In matrix form, the observed data can be compactly written as

$$\bar{Y} = \beta \bar{F}^\top + \bar{\mathcal{E}} + R, \quad (3.4)$$

for some “residual” matrix  $R$  capturing the error from trimming and the drift in  $Y_t$ .

Using the discrete observations and PCA analysis, we form estimates for the latent factors up to a rotation matrix. We denote

these estimates with  $\hat{f}_t$ . We then run OLS regression of  $\bar{g}_t$  on  $\hat{f}_t$  without intercept, and denote the estimated residuals with  $\hat{u}_t$ .

One can show (see [Theorem 2](#)) that, under our assumptions, we have the following convergence in probability

$$\frac{1}{n} \sum_{t=1}^n B_p \hat{f}_t \hat{u}_t w(t/n) \xrightarrow{\mathbb{P}} \int_0^T w(t/T) d\bar{H}' \langle f, u \rangle_t, \quad (3.5)$$

for some deterministic weight function  $w : [0, 1] \rightarrow \mathbb{R}$ , some diagonal matrix  $B_p$  with entries that are powers of  $p$ , and some invertible matrix  $\bar{H}$ . Obviously, if the null hypothesis is true and  $\langle f, u \rangle_t$  is identically zero on  $[0, T]$ , the limit in the above will be zero as well. However, this will not be in general the case when the null does not hold.

Our test consists of checking whether the limit in the above is statistically different from zero for a whole class of weight functions. In particular, we introduce the following family of weight functions:

$$\mathcal{W}_\eta(t) = \cos(\eta t) + \sin(\eta t), \text{ for } \eta \in [0, 2\pi] \text{ and } t \in [0, 1]. \quad (3.6)$$

Using Theorem 1 of Bierens and Ploberger (1997), we have that for a cadlag function  $f : [0, T] \rightarrow \mathbb{R}$ ,  $f(t) = 0$  almost everywhere on  $[0, T]$  if and only if  $\int_0^T \mathcal{W}_\eta(t/T) f(t) dt = 0$  for all  $\eta \in [0, 2\pi]$ .

This suggests the following test statistic for our null hypothesis:

$$\hat{\mathbb{S}} = \sum_{k=1}^K \sup_{\eta_k \in [0, 2\pi]} |\hat{\mathbb{S}}_k(\eta_k)|^2, \quad (3.7)$$

where

$$\hat{\mathbb{S}}_k(\eta_k) = \frac{1}{n} \sum_{t=1}^n \hat{f}_{t,k} \hat{u}_t \mathcal{W}_{\eta_k}(t/n), \quad k = 1, \dots, K. \quad (3.8)$$

### 3.2. Asymptotic Properties of the Test Statistic

For deriving the asymptotic properties of our statistic we need an assumption for the strength of the latent factors. To state this assumption, let

$$B_p = \text{diag}\{p^{\alpha_1/2}, \dots, p^{\alpha_K/2}\}, \quad 0 = \alpha_1 \leq \dots \leq \alpha_K < 1/2.$$

Our assumption for the strength of the factors is given by:

A1. Denote  $\Sigma_f = \langle f, f \rangle_T$  and assume that its eigenvalues are bounded away from zero and infinity. Furthermore, there is a  $K \times K$  diagonal matrix  $\bar{D}$ , whose diagonal elements are distinct and bounded away from zero and infinity, so that:

$$\left\| B_p \left( \Sigma_f^{1/2} \frac{1}{p} \beta' \beta \Sigma_f^{1/2} \right) B_p - \bar{D} \right\| = o_p(p^{-\alpha_K}), \text{ as } p \rightarrow \infty. \quad (3.9)$$

The rest of the assumptions are given in the Appendix. A few comments are in order for the above assumption. The standard case with strong factors corresponds to  $\alpha_1 = \dots = \alpha_K = 0$ . The power  $\alpha_j$  controls the strength of factor  $j$ , with higher values implying weaker factors. In this article, as in Bai and Ng (2023), weak factors mean weak factor loadings. In general, when factors are weak in the above-mentioned sense, inference for them is

harder, see the results in Bai and Ng (2023) for the case  $\alpha_K < 1/2$ . Under this condition, we can derive Central Limit Theorem (CLT) for our test statistic, provided that the factors are not very weak. When  $\alpha_K \geq 1/2$ , that is, when there are some very weak factors, even though these factors can still generate time-varying projection coefficients  $\gamma_t$ , their recovery is difficult and this seems to invalidate the CLT result below. Extending the results in the current article to cover presence of such weaker factors might still be possible but this is a rather nontrivial task. We leave such an extension for future work (also see Fan, Yan, and Zheng 2024; Choi and Yuan 2025 for studies of the case  $\alpha_K \geq 1/2$ ).

In the following theorem,  $\xrightarrow{\mathcal{L}-s}$  denotes stable convergence in law, see for example, Section VIII.5.c of Jacod and Shiryaev (2003). Our CLT is for the vector

$$\hat{\mathbb{S}}(\bar{\eta}) = (\hat{\mathbb{S}}_k(\eta_k) : k = 1, \dots, K), \quad \bar{\eta} = (\eta_1, \dots, \eta_K).$$

**Theorem 1.** Suppose assumptions A1–A3 hold. Let  $n, p \rightarrow \infty$  with

$$p^{2\alpha_K}/n \rightarrow 0, \quad n/p^{2-2\alpha_K} \rightarrow 0, \quad \text{and} \quad p^{\alpha_K} n^{1/2-2\tilde{\omega}} + p^{\alpha_K+1/2} n^{-1/2-\tilde{\omega}} \rightarrow 0, \quad (3.10)$$

for some  $\tilde{\omega} < \varpi$  and provided  $\varpi > 1/4$ . Then, under  $\mathbb{H}_0$  in (2.8):

$$\sqrt{n} B_p \hat{\mathbb{S}}(\bar{\eta}) \xrightarrow{\mathcal{L}-s} \mathbb{Z}_K(\bar{\eta}), \quad (3.11)$$

uniformly in  $\bar{\eta} \in [0, 2\pi]^K$ , and where  $\mathbb{Z}_K(\bar{\eta})$  is a  $K$ -dimensional vector-valued process, defined on an extension of the original probability space, which is  $\mathcal{F}$ -conditionally a zero-mean Gaussian process, with  $\mathcal{F}$ -conditional covariance kernel given in the Appendix.

We make several comments about the above limit result. First, we need a restriction on the growth of the dimension of the vector  $Y_t$  relative to the sampling frequency. This condition is given in (3.10) and is relatively weak when all of the factors are strong, that is, when  $\alpha_K = 0$ . In particular, the empirically relevant case in which  $p$  and  $n$  grow at the same rate is allowed for. The condition becomes naturally stronger when there are weaker factors. This is because the signal for the weak factors in  $Y$  is relatively low and thus various biases related to that extraction play more prominent role.

Second, the limit is mixed-Gaussian process with  $\mathcal{F}$ -conditional covariance kernel that is in general random. Third, the rate of convergence depends only on the sampling frequency and is determined by the martingale component of the observable factor that is orthogonal to the latent factors (in a martingale sense). One can show that the error in recovering the (rotated) latent factors is of higher asymptotic order and does not impact the limiting distribution in (3.11). We note, however, that the rotation matrix in presence of weak factors is shrinking asymptotically and this has an asymptotic effect on our statistic.

Finally, we note that the scaling of the statistic in (3.11) depends on the matrix  $B_p$ , which governs the strength of the latent factors (recall assumption A1 above). When the strength of the factors differs across the factors, then the elements of the vector  $\hat{\mathbb{S}}(\bar{\eta})$  converge at a different rate. In general,  $B_p$  will be

unknown. However, as we will see later, this will have no impact on the development of a feasible test on the basis of [Theorem 1](#).

We next present a convergence in probability result for the process  $\widehat{\mathbb{S}}(\bar{\eta})$ , which will allow us to study the power of the test.

**Theorem 2.** Suppose assumptions A1–A3 hold, let  $n, p \rightarrow \infty$ , and assume that the condition in (3.10) holds. Under both the null and alternative,

$$B_p \widehat{\mathbb{S}}(\bar{\eta}) \xrightarrow{\mathbb{P}} \frac{1}{T} \int_0^T \text{diag}(w_{\eta_1}(t/T), \dots, w_{\eta_K}(t/T)) \bar{H}' d\langle f, u \rangle_t, \quad (3.12)$$

uniformly in  $\bar{\eta} \in [0, 2\pi]^K$ , and where  $\bar{H}$  is some random  $K \times K$  invertible matrix.

The limit of  $\widehat{\mathbb{S}}(\bar{\eta})$  depends on a random invertible matrix  $\bar{H}$ . This is because the latent factors can be recovered only up to a rotation matrix. This, however, will not impact our test because our interest is only whether the limit in (3.12) is different from zero for any  $\bar{\eta}$ .

Similar to our CLT result in [Theorem 1](#), the convergence result in [Theorem 2](#) is about the scaled matrix  $B_p \widehat{\mathbb{S}}(\bar{\eta})$ . Weaker factors will lead to smaller in magnitude discrepancies, which will be naturally harder to detect.

### 3.3. Bootstrap Critical Values

For constructing a test on the basis of the limit result in [Theorem 1](#), we need feasible estimates of the covariance kernel. Instead of constructing such estimates, we will develop a bootstrap approach which is straightforward to implement. In the high-frequency setting, the factors typically exhibit nontrivial variation in their volatility. To account for that, therefore, we employ a wild bootstrap as in Mammen (1993). Specifically,

Step 1. Generate  $\eta_t^*$ ,  $t = 1, \dots, n$  as iid  $N(0, 1)$  random variables, and define

$$g_t^* = \widehat{\gamma}_0 + \widehat{\gamma}^\top \widehat{f}_t + \eta_t^* \widehat{u}_t.$$

Step 2. Let  $\widehat{\gamma}_1^*, \dots, \widehat{\gamma}_K^*$  denote the estimated slope coefficients when regressing  $g_t^*$  on  $\widehat{f}_t$ . Let  $\widehat{u}_t^*$  denote the corresponding time series residuals.

Step 3. Let

$$\widehat{\mathbb{S}}^* = \sum_{k=1}^K \sup_{\eta_k \in [0, 2\pi]} |\widehat{\mathbb{S}}_k^*(\eta_k)|^2, \quad (3.13)$$

where

$$\widehat{\mathbb{S}}_k^*(\eta) = \frac{1}{n} \sum_{t=1}^n \widehat{f}_{t,k} \widehat{u}_t^* \mathcal{W}_\eta(t/T), \quad k = 1, \dots, K. \quad (3.14)$$

Let

$$\widehat{\mathbb{S}}^*(\eta) = (\widehat{\mathbb{S}}_k^*(\eta) : k = 1, \dots, K).$$

The asymptotic behavior of the bootstrap statistic is given in the following theorem. In it, we denote  $\mathcal{F}$ -conditional convergence in law with  $\xrightarrow{\mathcal{L}|\mathcal{F}}$ , see Section VIII.5b of Jacod and Shiryaev (2003).

**Theorem 3.** Suppose assumptions A1–A3 hold, let  $n, p \rightarrow \infty$  and assume that the condition in (3.10) holds. Under both the null and alternative,

$$\sqrt{n} B_p \widehat{\mathbb{S}}^*(\bar{\eta}) \xrightarrow{\mathcal{L}|\mathcal{F}} \mathbb{Z}_K(\bar{\eta}), \quad (3.15)$$

uniformly in  $\bar{\eta} \in [0, 2\pi]^K$  and where  $\mathbb{Z}_K(\bar{\eta})$  is a process defined on an extension of the original probability space and which has the same  $\mathcal{F}$ -conditional law as the limiting process in [Theorem 1](#).

As seen from the result of the above theorem, the bootstrap statistic has the same  $\mathcal{F}$ -conditional limiting distribution as our test statistic. This allows for an easy way of constructing test of given asymptotic size as we show next.

### 3.4. Asymptotic Size and Power of the Test

For constructing the critical region of the test with asymptotic size of  $\alpha \in (0, 1)$ , we repeat steps 1–3 of the previous section  $B$  times and obtain  $\widehat{\mathbb{S}}_1^*, \dots, \widehat{\mathbb{S}}_B^*$ . We then compute the bootstrap  $p$ -value

$$p^* = \frac{1}{B} \sum_{b=1}^B 1\{\widehat{\mathbb{S}}_b^* + \widehat{\epsilon} \zeta_b^2/n > \widehat{\mathbb{S}} + \widehat{\epsilon} \zeta^2/n\}, \quad (3.16)$$

where  $\{\zeta_b\}_{b=1, \dots, B}$  and  $\zeta$  are standard normal variables, defined on an extension of the probability space, that are independent of  $\mathcal{F}$  and of each other,  $\widehat{\epsilon}$  is a consistent estimator, based on the available data, of some positive random variable  $\epsilon$ . The reason for adding a scaled and squared standard normal variable to our statistic is to ensure that the limiting distribution of the statistic has a probability density. It appears nontrivial to show that this is the case for the limiting distribution of  $\widehat{\mathbb{S}}$ . By adding an independent random variable to the test statistic, with a smooth probability density, we can ensure that this also holds for the limiting distribution of  $\widehat{\mathbb{S}} + \widehat{\epsilon} \zeta^2/n$  under the null hypothesis. In order to minimize the impact from doing this on the power of the test, one should pick  $\epsilon$  very small as we do in our empirical application, in which we set

$$\epsilon = 1/1000 \times \text{Var}(g_t).$$

We do this in a data-driven way in our numerical sections (and this is why we use  $\widehat{\epsilon}$  in defining  $p^*$ ).

The critical region  $\mathcal{C}_{n,\tau}$  of the test is then given by  $\mathcal{C}_{n,\tau} = \{p^* < \tau\}$ .

**Theorem 4.** Suppose assumptions A1–A4 hold, let  $n, p, B \rightarrow \infty$  and assume that the condition in (3.10) holds. We have

$$\mathbb{P}(\mathcal{C}_{n,\tau} | \mathbb{H}_0) \rightarrow \tau, \quad \mathbb{P}(\mathcal{C}_{n,\tau} | \mathbb{H}_1) \rightarrow 1. \quad (3.17)$$

Unlike [Theorems 1–3](#), which due to the presence of the scaling matrix  $B_p$ , contain infeasible limit results, the result in [Theorem 4](#) is feasible because the computation of the  $p$ -value in (3.16) does not need knowledge of the scaling matrix  $B_p$ . For asymptotic power one of the test, we need to make sure that the impact caused by time-variation in the projection coefficient  $\gamma_s$  due to the weakest latent factor on our statistic  $\widehat{\mathbb{S}}$  is bigger in asymptotic sense than the asymptotic size of  $\widehat{\mathbb{S}}$  under the null hypothesis. Naturally, presence of weaker factors reduces the power of the test but [Theorem 4](#) shows that for  $\alpha_K < 1/2$ , the weak factors do not affect the asymptotic power.

## 4. Monte Carlo Study

### 4.1. Setting

The baseline model for our Monte Carlo study is the one used in Liao and Todorov (2024), which is calibrated to match key properties of real asset return data like the one used in our empirical application. It is given by:

$$dY_{t,j} = \sigma_t \left( \beta_j^\top df_t + \bar{\sigma}_j dW_{t,j} \right), \quad j = 1, \dots, p, \quad (4.1)$$

where the univariate volatility process  $\sigma_t$  has the following square-root diffusion dynamics:

$$d\sigma_t^2 = 8.3(1 - \sigma_t^2)dt + \sigma_t dB_t, \quad (4.2)$$

with  $B_t$  being a standard Brownian motion that is independent from  $\{W_{t,j}\}_{j \geq 1}$ .

We generate the *baseline* factor and betas from:

$$\beta_j^0 \sim N(\mu_\beta, \Sigma_\beta), \text{ and}$$

$$df_t^0 = \Sigma_f^{1/2} dW_t, \text{ for } j = 1, \dots, p \text{ and } t \in [0, T],$$

for  $W_t$  being a three-dimensional Brownian motion independent of  $\{W_{t,j}\}_{j \geq 1}$  and  $B_t$ , and where

$$\Sigma_f = \text{diag}(1.198, 0.377, 0.264) \times 10^{-6}, \quad (4.3)$$

$$\mu_\beta = \begin{pmatrix} 0.8763 \\ 0.2818 \\ -0.2965 \end{pmatrix}, \quad \Sigma_\beta = \begin{pmatrix} 0.2326 & -0.2474 & 0.2603 \\ -0.2474 & 0.9224 & 0.0837 \\ 0.2603 & 0.0837 & 0.9139 \end{pmatrix}. \quad (4.4)$$

We consider three settings for the true factors and betas:

**Setting A:** Set the true betas and factors as the baseline:

$$\beta_i = \beta_i^0, \quad f_t = f_t^0, \quad i = 1, \dots, p, \quad t \in [0, T].$$

**Setting B:** The third factor is not active on the interval  $[0, T_0]$ , for some  $0 < T_0 < T$ :

$$\beta_i = \beta_i^0, \quad f_{t,3} = \begin{cases} 0, & t \in [0, T_0] \\ f_{t,3}^0, & t \in [T_0, T]. \end{cases}$$

**Setting C:** The loadings for the third factor have a structural break:

$$\beta_{i,j} = \beta_{i,j}^0, \quad j = 1, 2, \quad \beta_{i,t,3} = \begin{cases} \beta_{i,3}^0, & t \in [0, T/2] \\ \beta_{i,3}^1, & t \in [T/2, T]. \end{cases}, \quad f_t = f_t^0,$$

where

$$\beta_{i,3}^1 = \beta_{i,3}^0 + \tau \times z_i, \quad z_i \sim N(0, 0.9139),$$

and  $\tau$  is a parameter that determines the amount of change in the third column of  $\beta$ .

Finally, as in Liao and Todorov (2024), the scale of the idiosyncratic variance  $\bar{\sigma}_j$  is cross-sectionally iid and is drawn according to

$$\bar{\sigma}_j \sim \text{Uniform}([0.5, 1.5]) \times 1.1 \times 10^{-3}, \quad j = 1, \dots, p. \quad (4.5)$$

With such a choice for  $\bar{\sigma}_j$ , the share of idiosyncratic risk in total asset variance is around 40% for the median stock in the cross-section.

We turn next to the specification of the observable factor, which we model as

$$dg_t = \gamma_t^\top df_t + \sigma_g dW_t^g, \quad \sigma_g = 6.9 \times 10^{-4},$$

where  $W_t^g$  is a standard Brownian motion that is independent from  $W_t$ ,  $B_t$  and  $\{W_{t,j}\}_{j \geq 1}$ . The above value of  $\sigma_g$  is chosen so that the share of  $u_t$  in the variance of  $g_t$  is around 40%.  $\gamma_t$  is either constant (null hypothesis) or varies (alternative hypothesis):

**null:**  $\gamma_t$  stays constant:  $\gamma_t = \mu_\beta$ .

**alternative:**  $\gamma_t$  shifts by one at some point in the observation interval:

$$\gamma_t = \begin{cases} \mu_\beta, & t \in [0, T - T_1], \\ \mu_\beta + 1, & t \in [T - T_1, T], \end{cases}$$

for some  $0 < T_1 < T$  and where  $\mu_\beta$  is the mean of the latent factor loadings.

Our simulation setup, presented above, is rather general. In particular, we consider cases in which some of the latent factors are “active” over parts of the estimation window only or when the exposure to them changes discretely. As explained in Section 2, these cases are all covered by our analysis. In particular, for setting C, we can equivalently write the third element of the latent factor as  $f_{t,3} = f_{t,3}^0 + f_{t,3}^1$ , where

$$f_{t,3}^0 = \begin{cases} f_{t,3}, & t \in [0, T/2] \\ 0, & t \in [T/2, T]. \end{cases}, \quad f_{t,3}^1 = \begin{cases} 0, & t \in [0, T/2] \\ f_{t,3}, & t \in [T/2, T]. \end{cases}$$

This means that three-factor model with time-varying factor loadings in setting C is equivalent to a 4-factor model with constant loadings over the time interval  $[0, T]$ .

Finally, for the observation scheme, we use a cross-section of  $p = 500$  assets and assume that we sample asset prices either 80 or 40 times during a trading day. This corresponds approximately to sampling every five or 10 min in a 6.5 hr trading day. In all considered cases in the Monte Carlo, the time window is fixed to ten trading days.

### 4.2. Tuning Parameters

The tuning parameters of the test are set as follows. First, the truncation parameter is set in the following data-driven way:

$$\varpi = 0.49 \quad \text{and} \quad \gamma_j = 4 \times \sqrt{RV_{d,j}}, \quad j = 1, \dots, p, \quad (4.6)$$

where  $RV_{d,j}$  is the realized variance of asset  $j$  on the (trading) day  $d$  the increment belongs to, given by

$$RV_{d,j} = \sum_{i=\lfloor (d-1)/(252\Delta_n) \rfloor}^{\lfloor d/(252\Delta_n) \rfloor} (\Delta_i^n X_j)^2. \quad (4.7)$$

This way of setting truncation allows it to adapt to the level of volatility of the asset.

Next, for determining the number of latent factors, we follow Liao and Todorov (2024) and standardize the asset returns by estimates of their volatility in order to minimize the impact of idiosyncratic volatility. Then, the number of factors is estimated using the information criterion proposed by Bai and Ng (2002),

where we set  $g_{n,p} = \log\left(\frac{np}{n+p}\right)$  in the penalty term. Our test relies on a consistent estimator of the number of latent factors, and the above-mentioned way of determining  $K$  provides a consistent estimator for it, see Theorem B.1 in Liao and Todorov (2024). We conjecture, however, that our test continues to work (theoretically and empirically) even in cases where the number of factors is over-estimated in the limit, a case that was recently considered by Fan and Liao (2022).

Finally, the scaling estimator  $\hat{e}$  (recall our definition of  $p^*$  in (3.16)) is set 1/1000 times the square of the sample variance of the observable factor.

#### 4.3. Results

The Monte Carlo results are reported in Table 1. We can see from them that the test has overall good finite sample size properties. There is only a slight under-rejection in setting A for the coarser sampling frequency of 10 min and window size of one day only. Note, in particular, that the size of the test is close to the nominal level in settings B and C when either a factor is present only for part of the time or when factor loadings change. This shows the robustness of the test to such features in the data.

Table 1 reveals also good power properties of the test in general. Not surprisingly, the power increases when either the

length of the time window or the sampling frequency increases. Naturally, the harder to identify alternative from the data is the one in which the change in the relationship between observable and latent factors is present for a smaller fraction of the total sample.

Overall, Table 1 shows good finite sample properties of the developed test.

#### 5. Empirical Application

We use the techniques developed in the article to study the dynamic relation between latent risk factors embedded in the cross-section of stock returns and a market volatility factor. Earlier work, following Ang et al. (2006), has studied the pricing properties of market volatility after assuming temporal stability of the (linear) relation between market volatility and the systematic risk factors embedded in stock prices over nontrivial time horizons. We aim to test if this is the case empirically. We are not interested in this application in how volatilities of individual stocks relate in the cross-section. We note in this regard the recent work of Ding et al. (2025) who show that a relatively simple multiplicative model for individual stock volatility featuring a common volatility factor and an idiosyncratic one works well empirically. Our approach here can be also applied to study the

Table 1. Rejection probabilities in Monte Carlo.

|                      |                      | Settings |                     |                     |            |              |
|----------------------|----------------------|----------|---------------------|---------------------|------------|--------------|
|                      |                      | A        | B                   |                     | C          |              |
| Frequency            | Hypothesis           |          | $T_0 = \frac{T}{4}$ | $T_0 = \frac{T}{2}$ | $\tau = 1$ | $\tau = 0.1$ |
| Panel A: T = 1 day   |                      |          |                     |                     |            |              |
| 10 min               | Null                 | 0.022    | 0.044               | 0.056               | 0.048      | 0.038        |
|                      | Alter. $T_1 = T/10$  | 0.124    | 0.150               | 0.156               | 0.136      | 0.142        |
|                      | Alter. $T_1 = 3T/10$ | 0.720    | 0.716               | 0.696               | 0.662      | 0.722        |
| 5 min                | Null                 | 0.054    | 0.040               | 0.038               | 0.056      | 0.054        |
|                      | Alter. $T_1 = T/10$  | 0.164    | 0.196               | 0.226               | 0.176      | 0.196        |
|                      | Alter. $T_1 = 3T/10$ | 0.844    | 0.880               | 0.844               | 0.864      | 0.876        |
| Panel B: T = 5 days  |                      |          |                     |                     |            |              |
| 10 min               | Null                 | 0.050    | 0.054               | 0.072               | 0.052      | 0.046        |
|                      | Alter. $T_1 = T/10$  | 0.696    | 0.684               | 0.686               | 0.720      | 0.712        |
|                      | Alter. $T_1 = 3T/10$ | 1.000    | 1.000               | 1.000               | 1.000      | 1.000        |
| 5 min                | Null                 | 0.040    | 0.048               | 0.058               | 0.058      | 0.068        |
|                      | Alter. $T_1 = T/10$  | 0.836    | 0.862               | 0.826               | 0.856      | 0.866        |
|                      | Alter. $T_1 = 3T/10$ | 1.000    | 1.000               | 1.000               | 1.000      | 1.000        |
| Panel C: T = 10 days |                      |          |                     |                     |            |              |
| 10 min               | Null                 | 0.044    | 0.052               | 0.064               | 0.064      | 0.030        |
|                      | Alter. $T_1 = T/10$  | 0.958    | 0.964               | 0.954               | 0.956      | 0.968        |
|                      | Alter. $T_1 = 3T/10$ | 1.000    | 1.000               | 1.000               | 1.000      | 1.000        |
| 5 min                | Null                 | 0.038    | 0.044               | 0.050               | 0.046      | 0.048        |
|                      | Alter. $T_1 = T/10$  | 0.994    | 0.986               | 0.986               | 0.990      | 0.996        |
|                      | Alter. $T_1 = 3T/10$ | 1.000    | 1.000               | 1.000               | 1.000      | 1.000        |
| Panel D: T = 20 days |                      |          |                     |                     |            |              |
| 10 min               | Null                 | 0.046    | 0.044               | 0.054               | 0.066      | 0.040        |
|                      | Alter. $T_1 = T/10$  | 1.000    | 0.998               | 1.000               | 1.000      | 1.000        |
|                      | Alter. $T_1 = 3T/10$ | 1.000    | 1.000               | 1.000               | 1.000      | 1.000        |
| 5 min                | Null                 | 0.058    | 0.054               | 0.066               | 0.046      | 0.040        |
|                      | Alter. $T_1 = T/10$  | 1.000    | 1.000               | 1.000               | 1.000      | 1.000        |
|                      | Alter. $T_1 = 3T/10$ | 1.000    | 1.000               | 1.000               | 1.000      | 1.000        |

NOTE: The results are based on 500 Monte Carlo replications and number of bootstrap replications of  $B = 1000$ . The time window is 10 days. 5 and 10 min sampling frequencies correspond to 80 and 40 returns per day, respectively. The nominal size of the test is 5%.

common volatility factor and its relation with the cross-section of stock returns.

In our analysis we will use a measure of true market spot volatility that is constructed from options on the S&P 500 index by the CBOE options exchange that is less noisy than realized volatility and is robust to jumps. The ticker for this volatility index is SPOTVOL, and its construction is based on the theoretical analysis of Todorov (2019). Even though the volatility index is constructed from options, it is free of the risk premium embedded in them, see for example, Todorov (2019) and Todorov and Zhang (2022). The series is available at intraday frequency and this will allow us to study the dynamic interaction between market volatility factor and the latent systematic risk in stock prices over relatively short intervals of time. We plot the spot volatility index in Figure 1. As seen from the figure, our sample covers both periods of very high and very low volatility. The figure also reveals both high and low frequency variation in volatility.

The cross-section of stocks we use in our analysis changes on a yearly basis and comprises of the 500 largest stocks by market capitalization determined at the end of the previous calendar year. To alleviate potential concerns about the impact of market microstructure noise, we sample asset prices at the relatively coarse frequency of 10 min during each trading day. In addition to the cross-section of individual stocks, we will also use the SPY exchange traded fund (ETF) on the S&P 500 index as our proxy for the market portfolio.

We implement our test for temporal stability of the relation between the volatility factor and the latent ones by setting the tuning parameters exactly as in the Monte Carlo study. In Table 2, we report the results from performing the test for different levels of time aggregation, ranging from 1 day to 40 trading days. That is, for level of aggregation of  $D$  days, we split the sample in non-overlapping periods of length  $D$  days and then perform the test over these non-overlapping periods. The reported empirical rejection rates of the test in the second column of Table 2 correspond to the fraction of non-overlapping periods of length  $D$  days in our sample for which the test rejects the null hypothesis.

As seen from the reported results, there is nontrivial rejection of the null hypothesis of a constant linear relation between the volatility factor and the latent risk factors for long periods of aggregation. Indeed, the empirical rejection rate gets close to the nominal size of the test only when the time window is of

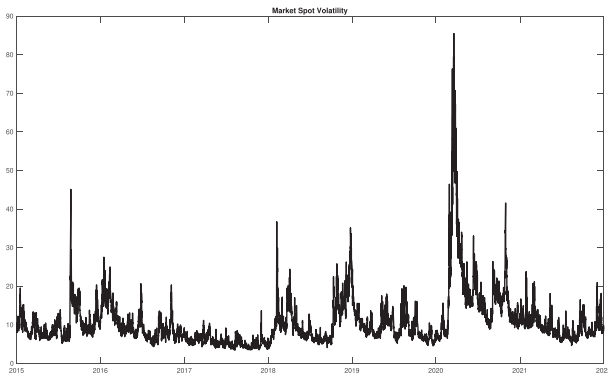


Figure 1. Market spot volatility.

Table 2. Empirical Results for Market Volatility

| Time aggregation | Empirical rejection Rate of Test | Average $R^2$ of regression | Average estimated number of factors |
|------------------|----------------------------------|-----------------------------|-------------------------------------|
| 1 day            | 0.0956                           | 0.4483                      | 4.1                                 |
| 5 days           | 0.2478                           | 0.4040                      | 6.1                                 |
| 10 days          | 0.3430                           | 0.3994                      | 7.0                                 |
| 20 days          | 0.4643                           | 0.4036                      | 8.5                                 |
| 40 days          | 0.6429                           | 0.4100                      | 9.5                                 |

NOTE: The nominal size of the test is 5%. The selection of the latent factors is done as in the Monte Carlo. Last column refers to the average estimated number of latent factors per period.

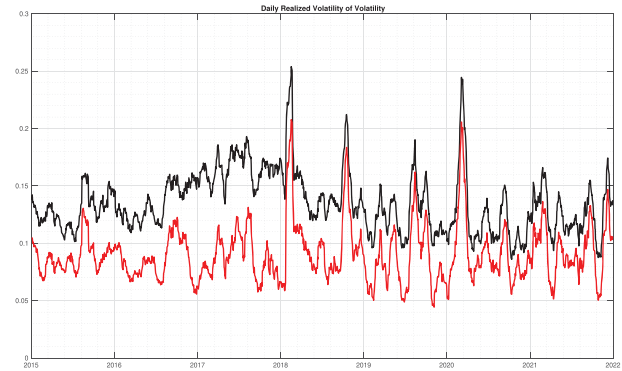


Figure 2. Market volatility of volatility. Black line corresponds to 20-day moving average of daily realized volatility of market volatility and red line is its counterpart for the daily linear projection of market volatility on the PCA factors.

length one day. This means that, in order to capture fully the risks in the market volatility factor, one has to estimate sensitivity toward volatility risk over much shorter periods than what has been done in earlier work.

The average  $R^2$  of the regressions, reported in Table 2, suggests that around 40% of the variation in the market spot volatility factor generates systematic moves in the cross-section of asset prices. The remaining part, which is rather nontrivial, is not relevant for the pricing of the assets in the cross section. Not accounting for this component of the volatility factor, as typically done in cross-sectional asset pricing, can lead to serious distortions when studying the pricing of volatility risk.

In the last column of Table 2, we report the average number of estimated latent factors per level of time aggregation. The number of estimated factors increases as we increase the level of time aggregation. One possible explanation for this is that the number of latent factors stays the same but the factor loadings change over the longer periods of aggregation. As explained in the Appendix (see also Monte Carlo setting C), such a dynamic factor model can be equivalently represented, within an augmented factor space, as a static factor model with some of the factors being dormant over parts of the period. Our theoretical setup covers such a scenario.

Finally, as a robustness check for the sensitivity of our analysis to the estimated number of factors, we performed the test with number of estimated latent factors increased by two from our original estimate of  $K$ . The empirical rejection rates of the test changed minimally relative to the ones reported in Table 2 when doing this. For brevity, we do not report these robustness results here.

In Figure 2, we plot the daily realized volatility of market volatility and the component of it that generates systematic risk

in stocks, that is, the linear projection on the latent systematic risk factors. We use daily aggregation when computing the linear projections. Even though market volatility changes a lot over our sample, we can see that market volatility of volatility does not exhibit significant low frequency variation. There are a few episodes with spikes in the overall volatility of volatility. Interestingly, when this happens, most of this volatility of volatility generates systematic risk in asset prices. Overall, the figure reveals that there is nontrivial variation in the importance of market volatility risk for the stocks.

Market volatility is well-known to be correlated with market risk. More specifically, shocks to the market portfolio and its volatility are known to be highly negatively correlated. Following Black (1976), this negative dependence is referred to as leverage effect. Given this strong dependence and the fact that the market factor is well-known to generate systematic risk in asset prices, a natural question is whether volatility generates systematic risk beyond the one due to the market risk factor. Put differently, can the component of volatility risk that generates systematic risk in the cross-section of asset prices be completely spanned by the market risk?

To answer this question we repeat the above analysis but for volatility risk that is locally orthogonal to market risk. We do this by estimating the linear projection of volatility returns on the market returns, and then looking at the residuals. To alleviate the impact due to the error from estimating the linear projection coefficients on the test, we use projection coefficients computed from the previous period when forming the estimates for volatility residuals over the current period. The test results are reported in Table 3.

These results mirror those for the volatility factor reported in Table 2. Indeed, the empirical rejection rate of the test is significantly above the nominal one for time windows above one trading day. This, of course, implies that the residual market volatility risk does generate systematic risk in the cross-section of assets. That said, the  $R^2$  of the regressions for the residual market volatility are significantly smaller than those for the market volatility itself. This means that, although present, the component of market volatility that is not spanned linearly by market risk and generates systematic risk in assets is relatively small.

Overall, our empirical analysis shows that market volatility generates systematic risk in asset prices which cannot be fully spanned by the market risk. To account for it, one should consider linear projections on the latent systematic risk factors embedded in the cross-section of stocks on relatively short time

windows of around one day. More generally, we expect that the observable risk factors will differ in terms of their relation with the latent ones embedded in the cross-section of stock returns. For some, which are more dynamic in nature, this relationship will hold only over very short periods while for others, for example, the market factor, this relationship will hold for much longer periods. These differences need to be accounted for when studying the cross-sectional pricing of the observable factors.

## 6. Conclusion

In this article we develop a nonparametric test for the temporal stability of a linear relation between observable factors and latent ones embedded in a high-dimensional vector obeying a linear factor model. The test uses high-frequency data of the observable factors and the high-dimensional vector. It evaluates the local behavior of residuals from global linear projections of the observed factors on the estimates of the latent ones. In an empirical application, the test is used to study the dynamic relation between the market volatility factor and the systematic risk in the cross-section of asset prices.

## Appendix A: Time-Varying Latent Factor Loadings

In this section we show that models with a finite number of changes in the latent factor loadings over the fixed time interval  $[0, T]$  plus additional small perturbations in them can be nested in our static factor model setup. Toward this end, suppose that  $Y_t$  has the following continuous-time dynamic latent factor model dynamics over  $[0, T]$ :

$$dY_t = \alpha_t dt + \tilde{\beta}_t d\tilde{f}_t + dJ_t + d\epsilon_t, \quad (\text{A.1})$$

where  $\alpha_t$ ,  $J_t$ , and  $\epsilon_t$  are the same as in (2.1) in the main text,  $\tilde{\beta}_t$  is a matrix of time-varying factor loadings and  $\tilde{f}_t$  is a vector of continuous (latent) systematic factors. The dimension of  $\tilde{f}_t$  is  $\tilde{K}$ .

We will further assume that  $\tilde{\beta}_t$  is a sum of piece-wise constant matrix-valued function over the time interval  $[0, T]$ , with finite number of changes within  $[0, T]$ , and a small perturbation. In this setting, there is an equivalent approximate static factor model representation for  $Y_t$  as we show now. To see this, suppose that  $\tilde{\beta}_t$  has the following representation

$$\tilde{\beta}_t = \tilde{\beta}_0 + \sum_{i=1}^m (\tilde{\beta}_{t_i} - \tilde{\beta}_{t_{i-1}}) 1(t < t_i) + r_t^\beta, \quad (\text{A.2})$$

where for some nonnegative integer  $m$ , we denote  $t_0 = 0 < t_1 < \dots < t_{m+1} = T$  and where  $r_t^\beta$  captures small variation in the latent factor loadings. Without loss of generality, we normalize  $r_t^\beta$  to be zero at time 0 and we make the following assumption for it: there exists a sequence of stopping times  $T_1, T_2, \dots$  increasing to infinity such that for  $s \leq t < T_m$ , we have

$$\begin{aligned} \sup_{j \geq 1, k \geq 1} \mathbb{E} |r_{t,j,k}^\beta|^q &< \infty, \quad \sup_{j \geq 1, k \geq 1} |\mathbb{E}_s(r_{t,j,k}^\beta - r_{s,j,k}^\beta)|^2 \\ &+ \sup_{j \geq 1, k \geq 1} \mathbb{E} |r_{t,j,k}^\beta - r_{s,j,k}^\beta|^2 \leq C_m |t - s|/n, \end{aligned} \quad (\text{A.3})$$

for  $r_{t,j,k}^\beta$  being the  $(j, k)$ th element of  $r_t^\beta$ , some sequence of positive constants  $C_m$ , and arbitrary  $q > 0$ . We recall that  $n$  denotes the number of times we sample over  $[0, T]$ . If the elements of  $r_t^\beta$  follow general Itô semimartingale dynamics, we have the above bounds but without the division by  $n$  in the second one. In this sense,  $r_t^\beta$  captures small

**Table 3.** Empirical results for residual market volatility.

| Time aggregation | Empirical rejection rate of test | Average $R^2$ of regression |
|------------------|----------------------------------|-----------------------------|
| 1 day            | 0.0845                           | 0.2417                      |
| 5 days           | 0.2588                           | 0.1122                      |
| 10 days          | 0.3455                           | 0.0841                      |
| 20 days          | 0.4286                           | 0.0890                      |
| 40 days          | 0.6571                           | 0.0957                      |

NOTE: Residual market volatility is the residual from linear projection of market volatility returns on market returns using linear projection coefficients estimated from the preceding period. The nominal size of the test is 5%. The selection of the latent factors is done as in the Monte Carlo.

perturbations in the factor loadings. The reason for dividing by  $n$  (and not by some other power of it) is that this choice guarantees that the time variation in  $\tilde{\beta}_t$  due to  $r_t^\beta$  does not have an asymptotic effect on our test.

The case  $m = 0$  and  $r_t^\beta = \mathbf{0}$ , for  $t \in [0, T]$ , corresponds to the classical static factor case commonly assumed in applications. For  $m > 0$ , we allow the factor loadings to change discretely over the interval  $[0, T]$ . The first component  $\beta_0$  is the baseline beta. The second component  $\sum_{i=1}^m (\tilde{\beta}_{t_i} - \tilde{\beta}_{t_{i-1}})1(t < t_i)$  identifies  $m$  number of “significant jumps” in the beta, occurring at times  $t_1, \dots, t_m$ .

We can alternatively write

$$\tilde{\beta}_t d\tilde{f}_t = \beta df_t + r_t^\beta d\tilde{f}_t, \text{ for } t \in [0, T], \quad (\text{A.4})$$

where

$$\beta = (\tilde{\beta}_0, \tilde{\beta}_{t_1} - \tilde{\beta}_{t_0}, \dots, \tilde{\beta}_{t_m} - \tilde{\beta}_{t_{m-1}}),$$

$$f_t = \begin{pmatrix} \tilde{f}_t \\ \tilde{f}_t 1(t \in (t_1, t_2]) \\ \vdots \\ \tilde{f}_t 1(t \in (t_{m-1}, t_m]) \end{pmatrix}, \quad (\text{A.5})$$

with  $\beta$  and  $f_t$  having dimensions  $p \times \tilde{K}(m+1)$  and  $\tilde{K}(m+1) \times 1$ , respectively. Thus, by expanding suitably the factor space, we can represent the dynamic factor model via the above approximate static factor model. The static factors  $f_t$  have a special structure if  $m \geq 1$ . Mainly, apart from the first one, they are active only on parts of the time interval  $[0, T]$ . This, and the fact that  $\tilde{f}_t$  exhibits jumps at the times  $\{t_i\}_{i=1}^m$ , does not matter for our inference procedures as our assumptions for the static factors accommodate such features (see assumption A2).

For the residual term due to the small perturbation in the factor loadings, we can denote

$$\bar{r}_t^\beta = \int_{(t-1)\Delta_n}^{t\Delta_n} r_s^\beta d\tilde{f}_s / \Delta_n, \quad t = 1, \dots, n, \quad (\text{A.6})$$

and

$$R^\beta = (\bar{r}_{t,j}^\beta; t = 1, \dots, n, j = 1, \dots, p), \quad (\text{A.7})$$

with  $\bar{r}_{t,j}^\beta$  being the  $j$ th element of  $\bar{r}_t^\beta$ . Given our assumption for  $r_t^\beta$  in (A.3) above, it is not hard to show that  $R^\beta$  is small and satisfies the conditions for the “residual” matrix  $R$  in Appendix D.1. In particular, the conditions in (D.5), (D.7), and (D.8) continue to hold with  $R$  replaced by  $R^\beta$ . Hence, all of the theoretical results in the article remain true in the more general setup in (A.1) when assumption (A.3) is imposed.

## Appendix B: Motivation for the Test for Finance Applications

The goal of this section is to motivate the importance of the test for possible applications in finance. In this section, as in our empirical application in the main text,  $Y_t$  will represent prices of individual stocks while  $g_t$  will be the value of an observable risk factor (e.g., market volatility as in our empirical application) whose price of risk we are trying to infer. Let us denote with  $M_t$  the value of the so-called stochastic discount factor, that is, an exponential martingale process that satisfies the following

$$P_t = \mathbb{E}_t(P_{t+s}M_{t+s})/M_t, \quad t, s > 0, \quad (\text{B.1})$$

and where  $P_t$  denotes the price of any financial asset at time  $t$ . For simplicity, we have set the risk-free interest rate to zero. If the latent factor  $f_t$  captures the priced risks in the cross-section of stocks as in the

classical Arbitrage Pricing Theory (APT), see, for example, Ross (1976) and Chamberlain and Rothschild (1983) as well as the conditional version of it (Hansen and Richard 1987), this means that  $\int_0^t \gamma_s^\top df_s$  captures the part of  $g_t$  that is relevant for asset pricing and  $\epsilon_t^g$  the part which is not. More specifically, suppose

$$\frac{dM_t}{M_t} = -\lambda_t^\top df_t, \quad (\text{B.2})$$

for some  $\lambda_t$  that determines the price of risk associated with the latent factors  $f_t$ . Then, in such a setting, the instantaneous price of risk for the observable factor  $g_t$  is given by

$$\lim_{\Delta \downarrow 0} \frac{\mathbb{E}_t(P_{t+\Delta} - P_t)}{P_t} = \lambda_t^\top c_t^{\text{ff}} \gamma_t, \quad (\text{B.3})$$

where we recall our notation of  $\gamma_t$  and  $c_t^{\text{ff}}$  in (2.4)–(2.5) in the main text.

Instead, if one were to assume that the projection coefficient  $\gamma_t$  remains constant on the interval  $[0, T]$ , then the estimate for the instantaneous price of risk for the observable factor will be based on estimating the quantity:

$$\lambda_t^\top c_t^{\text{ff}} \gamma, \quad (\text{B.4})$$

where the global projection coefficient  $\gamma$  is defined in (2.7) in the main text. As we explained in the main text, obtaining an estimate of  $\gamma$  is much easier than obtaining one of  $\gamma_t$ . Thus, from the point of view of estimation efficiency and/or robustness, we would like to use an estimator of the quantity in (B.4) but the latter is the true risk premium if and only if  $\gamma_t$  remains constant on  $[0, T]$ . This is what we test in this article.

## Supplementary Materials

The online supplement contains Appendix C of the article with the assumptions and Appendix D of the article with the proofs of the theoretical results.

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The authors report there are no competing interests to declare.

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